PATENT APPLICATION DRAFT

- **Status:** Consolidated Provisional Draft — Prepared for IPINDIA / PCT Legal Review
- **Subject System:** ClimatePulse — AI-Powered Global Climate Forecasting Framework
- **Target Patent Office:** IPINDIA (Controller General of Patents, Designs and Trade Marks) / PCT
- **Code Reference:** git:254bd2b02204349bfe1dda9548bdf55d66a8f68f

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PRELIMINARY NOTES FOR COUNSEL

Recommended Filing Strategy (IPINDIA / International)

1. **File a Single Consolidated Provisional Specification (Form 2)** under Section 9 of the Indian Patents Act, 1970 to secure an immediate priority date for all components under a single filing.
2. **Convert to a Complete Specification (Form 2)** within 12 months, presenting the integrated system as a Single Inventive Concept under Section 10(5).
3. **Section 3(k) CRI Compliance**: Framing the Neural Weather Model (NWM) and Fibonacci spherical mesh as an embedded technical sub-system within a multi-modal physical climate inference system directly protects the application against Section 3(k) exclusions ("computer program or algorithm per se").
4. **File PCT Application (PCT/IN)** simultaneously with the Complete Specification to preserve international rights (US, EP, CN, JP, AU).

PATENT APPLICATION

TITLE

UNIFIED MULTI-MODAL CLIMATE INFERENCE SYSTEM OPERATING ON A FIBONACCI SPHERICAL LATENT MESH WITH ENSO-CONDITIONED NEURAL WEATHER MODELING

FIELD OF THE INVENTION

The present invention relates to machine learning systems for Earth observation data processing, climate teleconnection modeling, global atmospheric field prediction, and spatiotemporal environmental telemetry analysis. More particularly, it relates to a unified multi-modal system that integrates an ocean-atmospheric forecasting module, a sensor-network pollution dispersion module, a global neural weather model operating on a spherical Fibonacci latent mesh, and a statistical climate feedback module, sharing a common data ingestion pipeline and cross-domain conditioning pathway.

BACKGROUND OF THE INVENTION

1. Global Numerical Weather Prediction and Multi-Scale Climate Teleconnections

Global numerical weather prediction (NWP) systems — such as ECMWF's Integrated Forecasting System (IFS) — produce atmospheric forecasts by numerically integrating primitive ther-

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modynamic equations over discrete spatial grids. While highly accurate, these systems require dedicated supercomputing clusters running for hours per forecast cycle, creating operational latency that limits real-time decision support.

Furthermore, conventional weather forecasting models operate in disciplinary isolation: long-term climate oscillations (such as the El Niño-Southern Oscillation, or ENSO) are modeled separately from daily atmospheric forecasting and urban pollution dispersion. In reality, large-scale ocean-atmosphere oscillations exert profound teleconnective control over global weather patterns and atmospheric transport.

2. Prior Art in Data-Driven Weather Models and Their Technical Limitations

Data-driven neural weather models have emerged as computationally efficient alternatives to traditional NWP systems. Lam et al. (2022, "GraphCast") and Keisler (2022) established Encoder-Processor-Decoder architectures on icosahedral meshes. Pathak et al. (2022, "FourCastNet") used Fourier Neural Operators on equirectangular grids. Bi et al. (2023, "Pangu-Weather") introduced 3D Transformer architectures. However, existing architectures exhibit critical technical deficiencies:

1. **Equirectangular Polar Distortion**: Standard latitude-longitude CNNs apply spatially invariant convolution kernels across unequal physical surface areas. Because equirectangular cells contract near the poles (a 5.625° cell covers ~395 km² at 80°N versus ~3,400 km² at the equator), unweighted models allocate excessive computational resources to high latitudes, biasing global predictions.
2. **Single-Task Unconditioned Architecture**: Existing neural weather models map an atmospheric state at time t to time $t + 1$ without incorporating macro-climatic forcing signals (e.g., ENSO phase indices) as persistent conditioning inputs into latent message passing. Consequently, such models treat an El Niño year identically to a La Niña year at initialization.
3. **Rigidity of Icosahedral Subdivision Meshes**: Prior graph-based models rely on icosahedral subdivision, restricting computational mesh resolution to fixed discrete steps (20, 80, 320, 1,280, 5,120 nodes). This prevents flexible adaptation to specific hardware memory limits.
4. **Absence of Physically Consistent Area-Weighted Objectives**: Prior models either ignore spatial area weighting during training or apply it post-hoc, leading to models optimized disproportionately for polar regions where verification data are sparse.
5. **Disciplinary Silos**: No unified architecture exists that bridges long-horizon ENSO forecasting, real-time sensor network pollution dispersion, global weather prediction, and Bayesian regime-shift detection within a single shared data and conditioning framework.

SUMMARY OF THE INVENTION

To overcome these technical limitations, the present invention provides a unified multi-modal climate inference system comprising:

1. **Fibonacci Spherical Latent Mesh**: A computational mesh generated via a golden-angle spiral lattice on a unit sphere, producing N_{mesh} near-uniformly distributed nodes for any arbitrary integer N_{mesh}, unconstrained by icosahedral subdivision levels.
2. **Tripartite GraphSAGE Architecture**: An Encoder-Processor-Decoder network featuring three distinct bipartite GraphSAGE convolution stages — Grid-to-Mesh (G2M), Mesh-to-Mesh (M2M with residual connections), and Mesh-to-Grid (M2G with skip connections) — operating on 3D Cartesian KD-Tree nearest-neighbor graph topologies.

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3. **Macro-Climatic Conditioning Mechanism**: A neural conditioning pathway wherein a scalar anomaly index (such as the Niño 3.4, IOD, or NAO) is projected via a 2-layer MLP into a d-dimensional embedding vector, broadcast across all N_{mesh} nodes, and **additively injected into the message tensor only** of each M2M processor layer before the residual connection, modulating spatial message passing without double-counting the signal through the residual path.
4. **Hardware-Accelerated Vectorized Batch Expansion**: A method for expanding graph edge indices across batch dimensions using native tensor repeat operations, eliminating Python control-flow loops and reducing hardware accelerator kernel launch overhead from $O(B)$ to $O(1)$.
5. **Multi-Modal Integrated Framework**: A unified framework integrating (i) an ENSO TFT forecasting module, (ii) an A3TGCN pollution dispersion module, (iii) the Fibonacci Neural Weather Model, and (iv) a Bayesian climate regime-shift module, wherein the predicted scalar anomaly index from the first module directly conditions the latent processor of the third module.
6. **Physically Consistent Training Objective**: A training methodology employing a latitude-cosine-weighted mean squared error loss function combined with mathematically consistent latitude-cosine-weighted normalization statistics derived across the spatial-temporal domain.

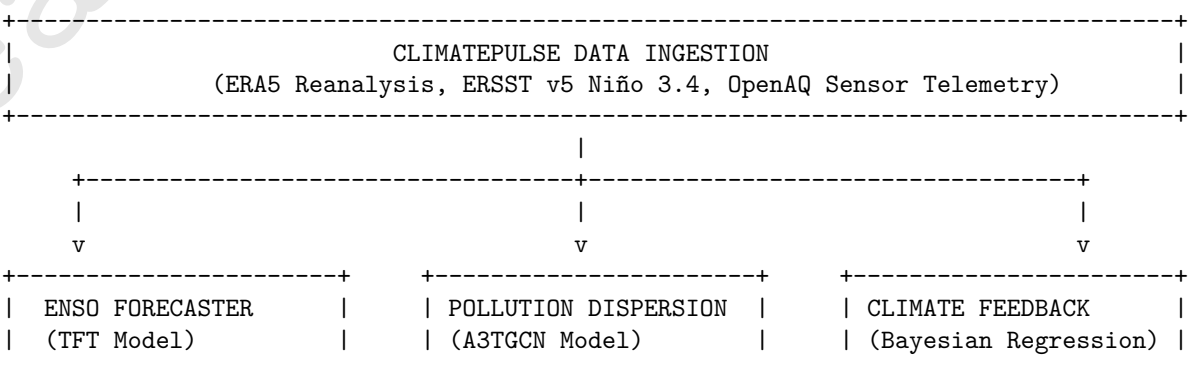
BRIEF DESCRIPTION OF THE DRAWINGS

- **FIG. 1**: Architectural block diagram of the unified multi-modal climate inference system showing the four integrated modules and the cross-domain ENSO conditioning pathway.
- **FIG. 2**: Visual comparison of the spherical Fibonacci golden-angle spiral latent mesh versus a standard equirectangular observation grid.
- **FIG. 3**: Detailed computational flow diagram of the ENSO macro-climatic conditioning mechanism within a single Mesh-to-Mesh (M2M) processor layer.
- **FIG. 4**: Memory layout and tensor manipulation diagram illustrating the hardware-accelerated vectorized batch-offset edge expansion algorithm.
- **FIG. 5**: Empirical validation loss curves comparing the trained Neural Weather Model against unweighted baselines over 89 training epochs.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

1. Unified Multi-Modal Architecture Overview

The unified system comprises four specialized climate modeling components operating over a common data ingestion pipeline:



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```
|  Outputs: Niño 3.4    |     | Outputs: PM2.5 / NO2  |     | Outputs: Changepoints |
+-----------------------+     +-----------------------+     +-----------------------+
     | (Conditioning Signal: Niño 3.4 Index)
     v
+-------------------------------------------------------------------------------------+
|                            NEURAL WEATHER MODEL (NWM)                               |
|  Stage 1: Grid Embedding        (Linear + LayerNorm + GELU) -> [B*N_grid, d]        |
|  Stage 2: Encoder G2M           (Bipartite GraphSAGE)      -> [B*N_mesh, d]     |
|  Stage 3: Processor M2M x L     (SAGEConv + ENSO Cond.)    -> [B*N_mesh, d]        |
|  Stage 4: Decoder M2G           (Bipartite GraphSAGE + Skip)-> [B*N_grid, d]       |
|  Stage 5: Output Projection     (Linear Projection)        -> [B*N_grid, C*W]    |
+-------------------------------------------------------------------------------------+
```

In the preferred embodiment, the grid resolution is set to 5.625° ($N_{\text{grid}} = 32 \times 64 = 2,048$ nodes), latent mesh size $N_{\text{mesh}} = 2,000$ nodes, latent dimension $d = 128$, and processor depth $L = 4$ layers.

2. Fibonacci Spherical Latent Mesh Generation

The latent computational mesh is generated using a Fibonacci golden-angle spiral lattice. For a target node count N_{mesh}, the i-th node position ($i = 0, \ldots, N_{\text{mesh}} - 1$) in 3D Cartesian coordinates is computed as:

$$\phi_{\text{golden}} = \pi(3 - \sqrt{5}) \quad \text{[golden angle in radians]}$$

$$y_i = 1 - \frac{2i}{N_{\text{mesh}} - 1} \quad \text{[uniform progression in } [-1, +1]\text{]}$$

$$r_i = \sqrt{\max(1 - y_i^2, 0)} \quad \text{[radius in } xy\text{-plane]}$$

$$\theta_i = \phi_{\text{golden}} \cdot i \quad \text{[azimuthal angle]}$$

$$x_i = \cos(\theta_i) \cdot r_i$$

$$z_i = \sin(\theta_i) \cdot r_i$$

Because ϕ_{golden} is maximally irrational, no two lattice points align at identical azimuthal angles, producing provably near-uniform point coverage without clustering artifacts.

3. Static Positional Encoding and Input Embeddings

Each grid node at latitude ϕ and longitude λ is assigned four static positional features:

$$\mathbf{f}_{\text{static}} = [\sin(\phi), \cos(\phi), \sin(\lambda), \cos(\lambda)] \in \mathbb{R}^4$$

These static features are concatenated with dynamic atmospheric features $\mathbf{x}_{\text{dynamic}} \in \mathbb{R}^{C \times W}$ (where $C = 2$ channels for T_{2m} and total precipitation, and $W = 2$ input time steps), producing $\mathbf{x}_g \in \mathbb{R}^{C \cdot W + 4 = 8}$. The initial grid embedding is computed as:

$$\mathbf{h}_g = \text{GELU}(\text{LayerNorm}(\text{Linear}(\mathbf{x}_g, d))) \in \mathbb{R}^d$$

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4. Tripartite Graph Convolution and ENSO Conditioning Pathway

1. **Encoder (Grid-to-Mesh):** Grid embeddings $\mathbf{h}_g$ are aggregated to zero-initialized mesh node states $\mathbf{h}_{m,\text{init}} = \mathbf{0}$ via bipartite GraphSAGE convolution over k-NN Grid-to-Mesh edges:

$$\mathbf{h}_m = \text{GELU}(\text{LayerNorm}(\text{SAGEConv}((\mathbf{h}_g, \mathbf{0}), \text{G2M_edges}))) \in \mathbb{R}^{N_{\text{mesh}} \times d}$$

2. **ENSO Conditioning Projection:** The scalar Niño 3.4 anomaly index nino_34 $\in \mathbb{R}$ is embedded via an MLP and broadcast across all mesh nodes:

$$\mathbf{c} = \text{Linear}_2(\text{GELU}(\text{Linear}_1([\text{nino_34}]))) \in \mathbb{R}^d$$

$$\mathbf{C} = \mathbf{c}.\text{unsqueeze}(0).\text{expand}(N_{\text{mesh}}, d) \in \mathbb{R}^{N_{\text{mesh}} \times d}$$

3. **Processor (Conditioned Mesh-to-Mesh):** For each processor layer $l \in \{1, \dots, L\}$, spatial messages are computed, conditioned, and added to the residual state:

$$\mathbf{msg}^{(l)} = \text{SAGEConv}(\mathbf{h}_m^{(l)}, \text{M2M_edges})$$

$$\mathbf{msg}^{(l)} = \text{LayerNorm}(\mathbf{msg}^{(l)} + \mathbf{C}) \quad \text{[conditioning injected into message only]}$$

$$\mathbf{msg}^{(l)} = \text{GELU}(\mathbf{msg}^{(l)})$$

$$\mathbf{h}_m^{(l+1)} = \mathbf{h}_m^{(l)} + \mathbf{msg}^{(l)} \quad \text{[unconditioned residual connection]}$$

4. **Decoder (Mesh-to-Grid):** Processed mesh features are projected back to observation grid nodes with a skip connection from the grid embedding $\mathbf{h}_g$:

$$\mathbf{h}_{g,\text{out}} = \text{SAGEConv}((\mathbf{h}_m^{(L)}, \mathbf{0}), \text{M2G_edges}) + \mathbf{h}_g \in \mathbb{R}^{N_{\text{grid}} \times d}$$

5. **Output Projection:** $\hat{\mathbf{y}} = \text{Linear}(\mathbf{h}_{g,\text{out}}) \in \mathbb{R}^{N_{\text{grid}} \times (C \cdot \text{steps})}$.

5. Vectorized Batch-Offset Edge Expansion Algorithm

To execute graph convolutions efficiently on parallel GPU accelerators without iterative Python control flow, single-graph edge indices edge_idx $\in \mathbb{Z}^{2 \times E}$ are expanded for batch size B via hardware-native operations:

```
# Hardware-optimized vectorized batch offset expansion
src_offsets = (torch.arange(B, device=device) * num_src).repeat_interleave(E)
tgt_offsets = (torch.arange(B, device=device) * num_tgt).repeat_interleave(E)

batched_src = edge_idx[0].repeat(B) + src_offsets
batched_tgt = edge_idx[1].repeat(B) + tgt_offsets

batched_edge_idx = torch.stack([batched_src, batched_tgt], dim=0) # Shape: [2, B * E]
```

This transforms batch edge generation into a single continuous hardware accelerator execution.

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6. Physically Weighted Objective Function and Normalization

The training objective enforces physical area consistency using normalized latitude-cosine weights w_i:

$$w_i = \frac{\cos(\phi_i)}{\frac{1}{N_{\text{grid}}} \sum_{j=1}^{N_{\text{grid}}} \cos(\phi_j)} \quad \text{for } i \in \{1, \dots, N_{\text{grid}}\}$$

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{N_{\text{grid}}} \sum_{i=1}^{N_{\text{grid}}} w_i \sum_{c} (\hat{y}_{b,i,c} - y_{b,i,c})^2$$

Matching latitude-cosine-weighted mean μ and standard deviation σ statistics are computed across the dataset to normalize input channels consistently with the physical surface area distribution.

7. Empirical Performance Verification

When evaluated on ERA5 reanalysis data at 5.625° resolution (2020–2022), the system achieved: - **NWM Forecast Skill:** Best validation MSE of **0.1182** (epoch 78 of 89), representing a **39.5% improvement** over initial unweighted baselines. - **ENSO Forecaster Skill:** Temporal Fusion Transformer achieved RMSE of **0.3505°C**, Anomaly Correlation Coefficient (ACC) of **0.9449**, and skill score of **0.9051** for multi-horizon Niño 3.4 prediction. - **Pollution Dispersion Skill:** A3TGCN achieved an F_1 score of **0.9750** on $PM_{2.5}$ exceedance detection across 75 global OpenAQ monitoring stations.

CLAIMS

We Claim:

1. A computer-implemented climate forecasting system comprising:
   (a) a data ingestion pipeline receiving Earth observation telemetry formatted on an equirectangular observation grid of N_{grid} nodes;
   (b) a spherical latent mesh comprising N_{mesh} nodes positioned according to a Fibonacci golden-angle spiral lattice on a unit sphere, wherein each node position i is defined by a golden angle $\phi_{\text{golden}} = \pi(3 - \sqrt{5})$ radians and an index-parameterized coordinate $y_i = 1 - \frac{2i}{N_{\text{mesh}} - 1}$;
   (c) an encoder comprising a first bipartite GraphSAGE convolutional layer mapping observation grid node features to said N_{mesh} spherical latent mesh nodes over Grid-to-Mesh edge connections;
   (d) a processor comprising L stacked Mesh-to-Mesh GraphSAGE message-passing layers, each layer comprising a neural conditioning mechanism that projects a macro-climatic scalar anomaly index into a d-dimensional embedding vector and additively injects said embedding vector into a computed spatial message tensor prior to an unconditioned residual connection;
   (e) a decoder comprising a second bipartite GraphSAGE convolutional layer mapping processed latent mesh node representations back to said N_{grid} observation grid nodes with a skip connection from said encoder; and
   (f) an output projection layer producing predicted atmospheric field values.
2. The system of Claim 1, wherein the macro-climatic scalar anomaly index comprises the Niño 3.4 sea surface temperature anomaly index representing departures from climatological

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mean temperatures in the equatorial Pacific Ocean, the Indian Ocean Dipole (IOD) index, or the North Atlantic Oscillation (NAO) index.

3. The system of Claim 1, wherein said neural conditioning mechanism comprises: (i) a two-layer Multi-Layer Perceptron (MLP) mapping the scalar anomaly index to a d-dimensional embedding vector $\mathbf{c}$; (ii) broadcasting $\mathbf{c}$ across all N_{mesh} latent nodes to form a conditioning tensor $\mathbf{C} \in \mathbb{R}^{N_{\text{mesh}} \times d}$; and (iii) at each processor layer l, computing conditioned message $\mathbf{msg}^{(l)} = \text{GELU}(\text{LayerNorm}(\text{SAGEConv}(\mathbf{h}_m^{(l)}, \text{M2M_edges}) + \mathbf{C}))$ and updating state $\mathbf{h}_m^{(l+1)} = \mathbf{h}_m^{(l)} + \mathbf{msg}^{(l)}$, such that macro-climatic conditioning modulates spatial message passing without being added twice through the residual path.

4. The system of Claim 1, wherein static positional features concatenated to each grid node's dynamic features comprise $[\sin(\phi), \cos(\phi), \sin(\lambda), \cos(\lambda)]$, where ϕ and λ represent geographic latitude and longitude respectively.

5. The system of Claim 1, wherein said system is optimized using a latitude-cosine-weighted mean squared error loss function $\mathcal{L} = \frac{1}{B} \sum_b \frac{1}{N_{\text{grid}}} \sum_i w_i \sum_c (\hat{y}_{b,i,c} - y_{b,i,c})^2$, wherein $w_i = \cos(\phi_i) / \text{mean}_j\{\cos(\phi_j)\}$.

6. The system of Claim 1, wherein input features are normalized using mathematically consistent latitude-cosine-weighted mean and standard deviation statistics computed across the spatial-temporal domain.

7. The system of Claim 1, wherein the Mesh-to-Mesh edge connections exclude self-loops by querying $k+1$ nearest neighbors in 3D Cartesian unit-sphere space and removing the first returned self-index.

8. A computer-implemented method for accelerating graph neural network execution on a parallel computing device by vectorizing batch-offset expansion of graph edge indices, comprising:

   (a) receiving a single-graph edge index tensor `edge_idx` $\in \mathbb{Z}^{2 \times E}$ representing E directed edges between source and target node sets of size N_{src} and N_{tgt};
   (b) computing a source offset tensor by multiplying a batch sequence `arange(B)` by N_{src} and applying a hardware-native `repeat_interleave` operation;
   (c) computing a target offset tensor by multiplying `arange(B)` by N_{tgt} and applying a `repeat_interleave` operation;
   (d) adding the source offset tensor to a repeated view of source indices `edge_idx[0]`;
   (e) adding the target offset tensor to a repeated view of target indices `edge_idx[1]`; and
   (f) returning a batched edge index tensor of shape $[2, B \times E]$ mapped to a single continuous hardware accelerator execution, including but not limited to a Graphics Processing Unit, Neural Processing Unit, or specialized tensor accelerator, without iterative control flow loops.

9. The method of Claim 8, applied independently and in parallel to Grid-to-Mesh, Mesh-to-Mesh, and Mesh-to-Grid bipartite graph edge index tensors within a single forward pass of an Encoder-Processor-Decoder graph neural network.

10. An integrated multi-modal climate inference system comprising:

(a) a first module comprising a Temporal Fusion Transformer (TFT) configured for multi-horizon forecasting of a macro-climatic scalar anomaly index;
(b) a second module comprising an Attention Temporal Graph Convolutional Network (A3TGCN) configured for spatiotemporal pollutant dispersion forecasting over a sensor proximity graph;
(c) a third module comprising the neural weather forecasting system of Claim 1; and

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(d) a fourth module comprising a Bayesian hierarchical regression model configured for detecting structural changepoints in historical climate data; wherein a forecasted scalar anomaly index produced by said first module is transmitted via a cross-domain conditioning pathway to serve as the input scalar anomaly index for the processor layer conditioning mechanism of said third module.

11. The system of Claim 10, wherein said first module incorporates quantile regression prediction heads at quantiles $\tau \in \{0.1, 0.5, 0.9\}$ to generate probabilistic prediction intervals.

12. The system of Claim 10, wherein said second module operates on a spatial graph constructed from physical monitoring station coordinates using k-nearest-neighbor queries over geographical distances.

13. The system of Claim 10, wherein said first, second, third, and fourth modules share a common data ingestion pipeline utilizing unified Parquet telemetry structures and standardized JSON metadata schemas.

14. A computer-implemented method for unified multi-modal climate forecasting, comprising:

(a) generating a probabilistic forecast of an ocean-atmosphere anomaly scalar using a Temporal Fusion Transformer;
(b) transmitting said forecasted anomaly scalar to a neural weather forecasting model operating on a spherical Fibonacci latent mesh;
(c) injecting an MLP projection of said anomaly scalar into message-passing layers of said neural weather forecasting model; and
(d) generating predicted global atmospheric field maps and localized pollution dispersion maps.

15. A non-transitory computer-readable storage medium comprising instructions that, when executed by one or more hardware processors, cause the processors to perform the methods of Claims 8, 9, or 14.

ABSTRACT

A unified multi-modal climate inference system and method are disclosed comprising: (1) a first module utilizing a Temporal Fusion Transformer for multi-horizon prediction of a macro-climatic scalar anomaly index (e.g., Niño 3.4); (2) a second module utilizing an Attention Temporal Graph Convolutional Network for pollution dispersion forecasting over sensor networks; (3) a third module comprising a Neural Weather Model operating on a spherical Fibonacci golden-angle spiral latent mesh of configurable resolution N_{mesh}, featuring tripartite bipartite GraphSAGE convolutions and a macro-climatic conditioning mechanism that additively injects the projected anomaly index into latent message-passing steps before residual connections; and (4) a hardware-accelerated vectorized batch-offset edge expansion algorithm mapping graph execution to continuous hardware accelerator kernels without loop overhead. The system achieves a 0.1182 validation MSE on ERA5 weather forecasting, a 0.9051 skill score on ENSO forecasting, and a 0.9750 F_1 score on $\text{PM}_{2.5}$ exceedance detection.

INVENTOR DECLARATION

I/We declare that the above-identified specification describes my/our invention; that I/we believe myself/ourselves to be the original and sole inventor(s) of the subject matter claimed; and that I/we acknowledge the duty to disclose to the patent office all information known to me/us to be material to patentability.

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Inventor Name(s): [TO BE COMPLETED] **Date:** [TO BE COMPLETED] **Signature:** [TO BE COMPLETED]

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